Event Horizon First PR Report

Generated: 2026-05-30T00:39:10.327Z

Executive Summary

Event Horizon now has a first playable vertical slice on `feat/first-playable`: a mobile-first Vite + PixiJS v8 canvas game where the player taps, swipes, and captures dark-energy orbs to delay a black-hole collapse. The run is deterministic from seed plus input timings, uses a fixed 60 Hz simulation step, records replay payloads, exports a three-frame share poster, and includes a minimal score-submit endpoint. The repo builds locally, unit tests pass, mobile Chrome smoke tests pass, and screenshots were captured from the actual running build.

Assumptions

- The repo was greenfield except for the initial commit, so a plain static Vite scaffold was chosen.
- The target playable slice is a prototype, not a balanced production game.
- `2026-05-29` is used as the alpha seed date because it was provided in the task context.
- GitHub Pages is the public static target at `/event-horizon/`; Netlify is supported with a root base path.
- A custom deterministic simulation is sufficient for radial gravity, tether capture, flybys, and shadow arms; no Matter.js or Planck.js dependency is justified yet.
- The Netlify endpoint validates and acknowledges replay payloads only; storage and anti-cheat are backlog items.

Chosen Stack and Why

- Vite + TypeScript: smallest viable static scaffold, fast local dev, and straightforward GitHub Pages base-path support.
- PixiJS v8: direct WebGL-preferred 2D rendering with a scene graph and generated textures. The implementation follows the PixiJS v8 async `Application.init()` pattern and `preference: 'webgl'` option from the official PixiJS docs.
- Custom simulation: deterministic, low-allocation motion was simpler than integrating a physics engine for this slice.
- Vitest: fast deterministic simulation and endpoint unit tests.
- Playwright with Chrome: local mobile/touch smoke checks for the target browser.
- Netlify Functions: minimal serverless score endpoint in-repo, matching Netlify's default `/.netlify/functions/<name>` routing.
- Google Apps Script sample: alternate deploy path using `doPost` and `ContentService` JSON output.

Sources used: [PixiJS Application docs](https://pixijs.com/8.x/guides/components/application), [PixiJS official skills repo](https://github.com/pixijs/pixijs-skills), [Netlify Functions docs](https://docs.netlify.com/functions/get-started/), [Google Apps Script Content Service docs](https://developers.google.com/apps-script/guides/content), and [Vite base config docs](https://main.vitejs.dev/config/shared-options).

Implementation Plan and Estimated Hours

1. Repo inspection and AGENTS.md: 0.25h
2. Vite + TypeScript + PixiJS scaffold: 0.75h
3. Deterministic simulation, fixed-step loop, RNG, and replay payloads: 2.0h
4. PixiJS scene, galaxy background, black hole, orbs, tethers, flybys, shadow arms, HUD: 2.0h
5. Tap/swipe input handler and logical viewport scaling: 1.0h
6. Collapse animation and posterizer export: 1.25h
7. Netlify score function and GAS sample: 0.75h
8. Unit, e2e, score endpoint, and artifact capture tests: 1.25h
9. README, PR body, PDF report, and final polish: 1.25h

Total estimate: 10.5h

Prioritized Backlog

1. Add durable score storage and abuse limits for Netlify or GAS.
2. Add replay playback UI and a deterministic replay verifier in CI.
3. Balance phase thresholds, orb spawn curves, and capture scoring through playtest data.

4. Add visual capture streak feedback and better end-run poster composition.
5. Add sound, haptics, and reduced-motion options.
6. Add GitHub Actions for build, lint, unit tests, and Playwright smoke tests.
7. Add accessibility affordances for non-touch desktop play.

Git and PR Commands

```
git switch -c feat/first-playable
npm install
npm run build
npm run lint
npm run test
npm run test:e2e
npm run score:test
npm run capture:artifacts
npm run report:pdf
git add .
git commit -m "Build first playable Event Horizon slice"
git push -u origin feat/first-playable
gh pr create --base main --head feat/first-playable --title "Build first playable Event Horizon slice" --body-file
docs/pr-body.md
```

File Tree

```
AGENTS.md
README.md
docs/artifacts/collapse-mobile.jpg
docs/artifacts/gameplay-mobile.jpg
docs/artifacts/share-poster.png
docs/first-pr-report.md
docs/pr-body.md
eslint.config.js
gas/score-submit.gs
index.html
netlify.toml
netlify/functions/score-submit.mjs
package-lock.json
package.json
playwright.config.ts
scripts/capture-artifacts.mjs
scripts/generate-report-pdf.mjs
scripts/test-score-submit.mjs
src/game/EventHorizonGame.ts
src/game/FixedStepLoop.ts
src/game/InputHandler.ts
src/game/Simulation.ts
src/game/constants.ts
src/game/math.ts
src/game/posterizer.ts
src/game/rng.ts
src/game/scoreClient.ts
src/game/types.ts
src/main.ts
src/styles.css
src/vite-env.d.ts
tests/e2e/playable.spec.ts
tests/mjs.d.ts
tests/score-submit.test.ts
tests/simulation.test.ts
tsconfig.json
vite.config.ts
```

Changed and Created File Notes

- `AGENTS.md`: practical repo instructions for future Codex work.
- `.gitignore`: ignores local OS, environment, build, coverage, and dependency artifacts.
- `README.md`: project overview, commands, assumptions, deployment, replay payload, and PR workflow.
- `index.html`: minimal app shell with canvas mount and compact restart/share controls.
- `src/main.ts`: bootstraps the game and exposes a small debug/test API.
- `src/styles.css`: fullscreen mobile-first canvas shell and compact icon controls.
- `src/game/EventHorizonGame.ts`: PixiJS scene, rendering, HUD, scaling, poster export, score submit call.
- `src/game/Simulation.ts`: deterministic gameplay state, phase changes, orbs, flybys, shadow arms, replay events.

- `src/game/FixedStepLoop.ts`: fixed 60 Hz simulation loop decoupled from rendering.
- `src/game/InputHandler.ts`: pointer/touch tap and swipe mapping to logical world coordinates.
- `src/game/rng.ts`: string hash and `mulberry32` seeded RNG.
- `src/game/posterizer.ts`: creates vertical share image from three gameplay frames.
- `src/game/scoreClient.ts`: posts replay payloads to the score endpoint.
- `src/game/constants.ts`, `math.ts`, `types.ts`, `vite-env.d.ts`: shared constants, helpers, and types.
- `netlify/functions/score-submit.mjs`: minimal score validation endpoint.
- `gas/score-submit.gs`: Google Apps Script `doPost` JSON endpoint sample.
- `netlify.toml`: Netlify build, publish, and functions directory config.
- `package.json`, `package-lock.json`: dependencies and runnable scripts.
- `tsconfig.json`, `vite.config.ts`, `eslint.config.js`, `playwright.config.ts`: TypeScript, Vite, lint, and browser test config.
- `tests/**/*.ts`: deterministic replay, score endpoint, and mobile Chrome smoke tests.
- `scripts/capture-artifacts.mjs`: captures mobile gameplay, poster, and collapse images.
- `scripts/test-score-submit.mjs`: standalone endpoint sanity test.
- `scripts/generate-report-pdf.mjs`: generates the requested PDF artifact.
- `docs/artifacts/*`: screenshots and poster image from the actual build.
- `docs/pr-body.md`: PR description source.

Major Code Snippets

`index.html`

```
<main id="app" aria-label="Event Horizon game">
  <div id="game-shell">
    <div id="game-root" aria-label="Playable canvas"></div>
    <div id="status-layer" aria-live="polite">
      <button id="restart-button" type="button" aria-label="Restart run" title="Restart run">!»</button>
      <button id="share-button" type="button" aria-label="Create share poster" title="Create share poster">!ê</button>
      <a id="poster-link" download="event-horizon-poster.png" aria-label="Download share poster">!é</a>
    </div>
  </div>
</main>
<script type="module" src="/src/main.ts"></script>
```

Main Entry

```
const game = new EventHorizonGame(root, {
  seed: 'eh-2026-05-29-alpha',
  startedAt: 1780051200000
});

await game.start();
```

Fixed-Step Loop

```
while (this.accumulatorMs >= FIXED_STEP_MS) {
  this.step(FIXED_STEP_MS);
  this.accumulatorMs -= FIXED_STEP_MS;
}

this.render(this.accumulatorMs / FIXED_STEP_MS);
```

`mulberry32`

```
export function mulberry32(seed: number): RandomSource {
  let state = seed >>> 0;
  return () => {
    state = (state + 0x6d2b79f5) >>> 0;
    let next = state;
    next = Math.imul(next ^ (next >>> 15), next | 1);
    next ^= next + Math.imul(next ^ (next >>> 7), next | 61);
    return ((next ^ (next >>> 14)) >>> 0) / 4294967296;
  };
}
```

PixiJS Init

```
await this.app.init({
  autoDensity: true,
  autoStart: false,
  backgroundAlpha: 0,
  preference: 'webgl',
```

```
  preserveDrawingBuffer: true,
  powerPreference: 'high-performance',
  resizeTo: this.root,
  resolution: Math.min(window.devicePixelRatio || 1, 2)
});
```

Input Handler

```
if (moved >= this.swipeThresholdSq) {
  this.callbacks.onSwipe(this.start, end);
} else {
  this.callbacks.onTap(end);
}
```

Posterizer

```
const loadedFrames = await Promise.all(frames.slice(-3).map((frame) => loadImage(frame.dataUrl)));
context.fillText(`survived ${formatTime(stats.survivalMs)}  •  phase ${stats.phase}`, 52, footerY + 154);
return canvas.toDataURL('image/png', 0.86);
```

GAS `doPost`

```
function doPost(e) {
  var payload = JSON.parse(e.postData.contents || '{}');
  return ContentService.createTextOutput(JSON.stringify({
    ok: true,
    acceptedAt: new Date().toISOString(),
    score: Number(payload.score || 0),
    survivalMs: Number(payload.survivalMs || 0)
  })).setMimeType(ContentService.MimeType.JSON);
}
```

Sample Replay Payload


```
{
  "version": 1,
  "seed": "eh-2026-05-29-alpha",
  "startedAt": 1780051200000,
  "survivalMs": 67421,
  "score": 1280,
  "energyCaptured": 87,
  "maxStreak": 24,
  "tapEvents": [],
  "swipeEvents": [],
  "phaseTransitions": []
}
```


Test Results

- `npm run build`: passed. TypeScript compiled and Vite produced `dist/`.
- `npm run lint`: passed.
- `npm run test`: passed, 2 files and 5 tests.
- `npm run test:e2e`: passed, 4 mobile Chrome tests.
- `npm run score:test`: passed, score endpoint returned status `200` and `{ "ok": true }`.
- Local Chrome smoke test: passed via Playwright and in-app browser visual inspection.
- Deterministic replay sanity: passed. Same seed plus same input timings produced identical replay payloads.
- Touch simulation sanity: passed. Tap and swipe events are captured in replay payloads.
- Posterizer export: passed. Browser test returns a `data:image/png;base64,...` poster.
- Score submit test: passed against `/.netlify/functions/score-submit` handler import.

Performance Notes

- The simulation uses a fixed 60 Hz step and avoids allocating per-frame input or entity arrays.
- Orb and flyby sprites use generated textures rather than drawing new art every frame.
- Background art is drawn once; per-frame `Graphics` clearing is limited to tethers, arms, HUD fill, and flyby trails.
- No heavy filters or physics engine are used.
- `preserveDrawingBuffer` supports poster screenshots but can cost some GPU performance; if poster export changes later, replacing it with targeted render-texture capture may be better.
- The main visible bottleneck risk is `canvas.toDataURL()` during poster capture; it is user-triggered rather than hot-loop work.

Deployment Notes

Netlify

- Build command: `VITE_BASE_PATH=/ npm run build`
- Publish directory: `dist`
- Functions directory: `netlify/functions`
- Score function route: `/.netlify/functions/score-submit`
- Branch deploys: connect the GitHub repo in Netlify and enable branch deploys for PR previews.
- Local Netlify dev: `npx netlify dev` from the repo root.

GitHub Pages

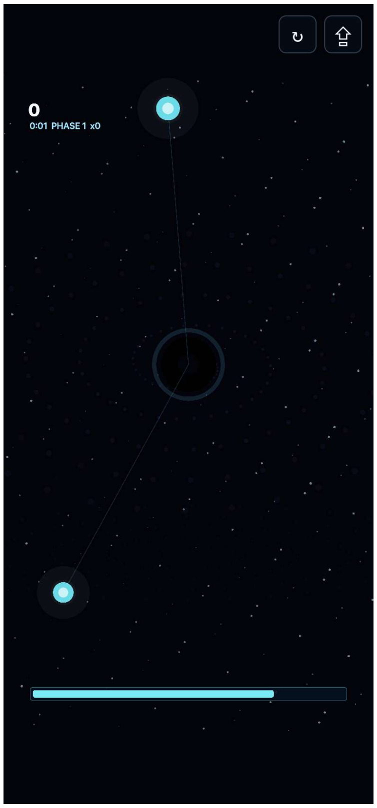
- Default Vite `base` is `/event-horizon/`.
- Static build command: `npm run build`
- Publish the generated `dist/` content through the chosen GitHub Pages workflow.
- Environment assumption: GitHub Pages uses a subpath, while Netlify uses root. Override with `VITE_BASE_PATH=/` for root deploys.

PR Summary

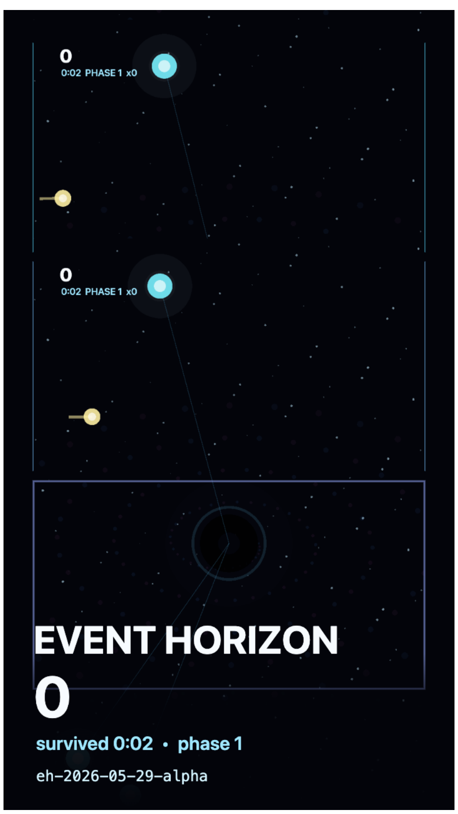
This PR introduces the first playable Event Horizon slice: deterministic simulation, PixiJS rendering, one-thumb input, replay payloads, phase-based collapse pressure, score submission samples, tests, documentation, screenshots, and this PDF report.

Screenshots and Poster

Gameplay mobile screenshot



Three-frame share poster



Collapse mobile screenshot

File Diffs

Diff Stat

```
 .gitignore                         |    9 +
 AGENTS.md                          |    9 +
 README.md                          |  105 +
 docs/artifacts/collapse-mobile.jpg |  Bin 0 -> 31259 bytes
 docs/artifacts/gameplay-mobile.jpg |  Bin 0 -> 24555 bytes
 docs/artifacts/share-poster.png    |  Bin 0 -> 130509 bytes
 docs/first-pr-report.md            |  299 +++
 docs/pr-body.md                    |   19 +
 eslint.config.js                   |   37 +
 gas/score-submit.gs                |   29 +
 index.html                         |   23 +
 netlify.toml                       |   12 +
 netlify/functions/score-submit.mjs |   67 +
 package-lock.json                  | 3970 ++++++++++++++++++++++++++++++++++++
 package.json                       |   30 +
 playwright.config.ts               |   31 +
 scripts/capture-artifacts.mjs      |   55 +
 scripts/generate-report-pdf.mjs    |  180 ++
 scripts/test-score-submit.mjs      |   30 +
 src/game/EventHorizonGame.ts       |  475 +++++
 src/game/FixedStepLoop.ts          |   54 +
 src/game/InputHandler.ts           |   79 +
 src/game/Simulation.ts             |  553 +++++
 src/game/constants.ts              |   11 +
 src/game/math.ts                   |   42 +
 src/game/posterizer.ts             |  110 +
 src/game/rng.ts                    |   25 +
 src/game/scoreClient.ts            |   35 +
 src/game/types.ts                  |   90 +
 src/main.ts                        |   48 +
 src/styles.css                     |   92 +
 src/vite-env.d.ts                  |    9 +
 tests/e2e/playable.spec.ts         |   48 +
 tests/mjs.d.ts                     |    4 +
 tests/score-submit.test.ts         |   40 +
 tests/simulation.test.ts           |   41 +
 tsconfig.json                      |   20 +
 vite.config.ts                     |   21 +
 38 files changed, 6702 insertions(+)
```

Unified Diff

```
diff --git a/.gitignore b/.gitignore
new file mode 100644
index 0000000..11553ed
--- /dev/null
+++ b/.gitignore
@@ -0,0 +1,9 @@
+node_modules/
+dist/
+coverage/
+playwright-report/
+test-results/
+.DS_Store
+.env
+.env.*
+!.env.example
diff --git a/AGENTS.md b/AGENTS.md
new file mode 100644
index 0000000..6de8651
--- /dev/null
+++ b/AGENTS.md
@@ -0,0 +1,9 @@
+# Event Horizon Agent Notes
+
+- Work from the repo root and keep changes scoped to the playable slice.
+- Use `npm run dev`, `npm run build`, `npm run test`, and `npm run lint` as the main verification commands.
+- Keep the game simulation deterministic: fixed 60 Hz steps, seeded `mulberry32`, and input events recorded with simulation
timestamps.
+- Treat PixiJS as the rendering layer only. Keep game state and replay logic in framework-light TypeScript modules.
+- The logical playfield is `1080 x 1920` portrait and must scale to the viewport without distortion.
+- GitHub Pages builds should use the `/event-horizon/` base path. Netlify builds should set `VITE_BASE_PATH=/`.
+- Avoid committing generated build output, local OS files, credentials, or large binary artifacts other than the requested PDF
report.
diff --git a/README.md b/README.md
new file mode 100644
index 0000000..a7f74e9
--- /dev/null
+++ b/README.md
@@ -0,0 +1,105 @@
+# Event Horizon
+
+Event Horizon is a mobile-first, one-thumb casual survival prototype about delaying a galaxy's collapse into a black hole. The
first vertical slice is a plain Vite + PixiJS v8 site with deterministic simulation state kept outside the renderer.
+
+## Current Slice
```

```
+
+- Logical portrait playfield: `1080 x 1920`
+- PixiJS canvas renderer with WebGL preference
+- Fixed `60 Hz` simulation step decoupled from render frames
+- Seeded `mulberry32` RNG and replay payloads from seed + input timings
+- Tap and swipe input recording
+- Dark-energy orbs, tether capture, flyby bonuses, and shadow-arm hazards
+- Phase-based black-hole visibility and gravity pressure
+- Bottom dark-energy meter, score/time HUD, and collapse end animation
+- Share poster export from three captured gameplay frames
+- Minimal Netlify score submit function plus Google Apps Script sample
+
+## Assumptions
+
+- The initial repo was empty, so this is a greenfield Vite scaffold.
+- Custom deterministic motion is enough for the vertical slice; no Matter.js or Planck.js dependency is needed yet.
+- GitHub Pages is the public static target and uses `/event-horizon/` as the base path.
+- Netlify deploys should set `VITE_BASE_PATH=/` through the included `netlify.toml`.
+- The score function validates and acknowledges payloads only; durable storage is a backlog item.
+
+## Commands
+
+```bash
+npm install
+npm run dev
+npm run build
+npm run lint
+npm run test
+npm run test:e2e
+npm run score:test
+npm run report:pdf
+```
+
+Local Vite serves the game at:
+
+```text
+http://127.0.0.1:5173/event-horizon/
+```
+
+## Deployment
+
+### GitHub Pages
+
+The default Vite base path is `/event-horizon/`, so `npm run build` produces static assets compatible with:
+
+```text
+https://kevinhegg.github.io/event-horizon/
+```
+
+### Netlify
+
+`netlify.toml` uses:
+
+```toml
+[build]
+  command = "VITE_BASE_PATH=/ npm run build"
+  publish = "dist"
+
+[functions]
+  directory = "netlify/functions"
+```
+
+The score endpoint is available at:
+
+```text
+/.netlify/functions/score-submit
+```
+
+For local Netlify function routing, run Netlify CLI from the repo root:
+
+```bash
+npx netlify dev
+```
+
+## Replay Payload Shape
+
+```json
+{
+  "version": 1,
+  "seed": "eh-2026-05-29-alpha",
+  "startedAt": 1780051200000,
+  "survivalMs": 67421,
+  "score": 1280,
+  "energyCaptured": 87,
+  "maxStreak": 24,
+  "tapEvents": [],
+  "swipeEvents": [],
+  "phaseTransitions": []
+}
+```
+
+## First PR Workflow
```

````diff
+
+```bash
+git switch -c feat/first-playable
+git add .
+git commit -m "Build first playable Event Horizon slice"
+git push -u origin feat/first-playable
+gh pr create --base main --head feat/first-playable --title "Build first playable Event Horizon slice" --body-file docs/pr-body.md
+```
diff --git a/docs/first-pr-report.md b/docs/first-pr-report.md
new file mode 100644
index 0000000..2f7e42c
--- /dev/null
+++ b/docs/first-pr-report.md
@@ -0,0 +1,299 @@
+# Executive Summary
+
+Event Horizon now has a first playable vertical slice on `feat/first-playable`: a mobile-first Vite + PixiJS v8 canvas game where the player taps, swipes, and captures dark-energy orbs to delay a black-hole collapse. The run is deterministic from seed plus input timings, uses a fixed 60 Hz simulation step, records replay payloads, exports a three-frame share poster, and includes a minimal score-submit endpoint. The repo builds locally, unit tests pass, mobile Chrome smoke tests pass, and screenshots were captured from the actual running build.
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+## Assumptions
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+- The repo was greenfield except for the initial commit, so a plain static Vite scaffold was chosen.
+- The target playable slice is a prototype, not a balanced production game.
+- `2026-05-29` is used as the alpha seed date because it was provided in the task context.
+- GitHub Pages is the public static target at `/event-horizon/`; Netlify is supported with a root base path.
+- A custom deterministic simulation is sufficient for radial gravity, tether capture, flybys, and shadow arms; no Matter.js or Planck.js dependency is justified yet.
+- The Netlify endpoint validates and acknowledges replay payloads only; storage and anti-cheat are backlog items.
+
+## Chosen Stack and Why
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+- Vite + TypeScript: smallest viable static scaffold, fast local dev, and straightforward GitHub Pages base-path support.
+- PixiJS v8: direct WebGL-preferred 2D rendering with a scene graph and generated textures. The implementation follows the PixiJS v8 async `Application.init()` pattern and `preference: 'webgl'` option from the official PixiJS docs.
+- Custom simulation: deterministic, low-allocation motion was simpler than integrating a physics engine for this slice.
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+
+Sources used: [PixiJS Application docs](https://pixijs.com/8.x/guides/components/application), [PixiJS official skills repo](https://github.com/pixijs/pixijs-skills), [Netlify Functions docs](https://docs.netlify.com/functions/get-started/), [Google Apps Script Content Service docs](https://developers.google.com/apps-script/guides/content), and [Vite base config docs](https://main.vitejs.dev/config/shared-options).
+
+## Implementation Plan and Estimated Hours
+
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+9. README, PR body, PDF report, and final polish: 1.25h
+
+Total estimate: 10.5h
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+## Prioritized Backlog
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+```
+
+## File Tree
+
````

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+```text
+AGENTS.md
+README.md
+docs/artifacts/collapse-mobile.jpg
+docs/artifacts/gameplay-mobile.jpg
+docs/artifacts/share-poster.png
+docs/first-pr-report.md
+docs/pr-body.md
+eslint.config.js
+gas/score-submit.gs
+index.html
+netlify.toml
+netlify/functions/score-submit.mjs
+package-lock.json
+package.json
+playwright.config.ts
+scripts/capture-artifacts.mjs
+scripts/generate-report-pdf.mjs
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+src/game/EventHorizonGame.ts
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+src/game/scoreClient.ts
+src/game/types.ts
+src/main.ts
+src/styles.css
+src/vite-env.d.ts
+tests/e2e/playable.spec.ts
+tests/mjs.d.ts
+tests/score-submit.test.ts
+tests/simulation.test.ts
+tsconfig.json
+vite.config.ts
+```
+
+## Changed and Created File Notes
+
+- `AGENTS.md`: practical repo instructions for future Codex work.
+- `.gitignore`: ignores local OS, environment, build, coverage, and dependency artifacts.
+- `README.md`: project overview, commands, assumptions, deployment, replay payload, and PR workflow.
+- `index.html`: minimal app shell with canvas mount and compact restart/share controls.
+- `src/main.ts`: bootstraps the game and exposes a small debug/test API.
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+- `src/game/rng.ts`: string hash and `mulberry32` seeded RNG.
+- `src/game/posterizer.ts`: creates vertical share image from three gameplay frames.
+- `src/game/scoreClient.ts`: posts replay payloads to the score endpoint.
+- `src/game/constants.ts`, `math.ts`, `types.ts`, `vite-env.d.ts`: shared constants, helpers, and types.
+- `netlify/functions/score-submit.mjs`: minimal score validation endpoint.
+- `gas/score-submit.gs`: Google Apps Script `doPost` JSON endpoint sample.
+- `netlify.toml`: Netlify build, publish, and functions directory config.
+- `package.json`, `package-lock.json`: dependencies and runnable scripts.
+- `tsconfig.json`, `vite.config.ts`, `eslint.config.js`, `playwright.config.ts`: TypeScript, Vite, lint, and browser test config.
+- `tests/**/*.ts`: deterministic replay, score endpoint, and mobile Chrome smoke tests.
+- `scripts/capture-artifacts.mjs`: captures mobile gameplay, poster, and collapse images.
+- `scripts/test-score-submit.mjs`: standalone endpoint sanity test.
+- `scripts/generate-report-pdf.mjs`: generates the requested PDF artifact.
+- `docs/artifacts/*`: screenshots and poster image from the actual build.
+- `docs/pr-body.md`: PR description source.
+
+## Major Code Snippets
+
+### `index.html`
+
+```html
+<main id="app" aria-label="Event Horizon game">
+  <div id="game-shell">
+    <div id="game-root" aria-label="Playable canvas"></div>
+    <div id="status-layer" aria-live="polite">
+      <button id="restart-button" type="button" aria-label="Restart run" title="Restart run">!»</button>
+      <button id="share-button" type="button" aria-label="Create share poster" title="Create share poster">!ê</button>
+      <a id="poster-link" download="event-horizon-poster.png" aria-label="Download share poster">!é</a>
+    </div>
+  </div>
+</main>
+<script type="module" src="/src/main.ts"></script>
+```
+
+### Main Entry
+
+```ts
+const game = new EventHorizonGame(root, {
+  seed: 'eh-2026-05-29-alpha',
+  startedAt: 1780051200000
````

````
+});
+
+await game.start();
+```
+
+### Fixed-Step Loop
+
+```ts
+while (this.accumulatorMs >= FIXED_STEP_MS) {
+  this.step(FIXED_STEP_MS);
+  this.accumulatorMs -= FIXED_STEP_MS;
+}
+
+this.render(this.accumulatorMs / FIXED_STEP_MS);
+```
+
+### `mulberry32`
+
+```ts
+export function mulberry32(seed: number): RandomSource {
+  let state = seed >>> 0;
+  return () => {
+    state = (state + 0x6d2b79f5) >>> 0;
+    let next = state;
+    next = Math.imul(next ^ (next >>> 15), next | 1);
+    next ^= next + Math.imul(next ^ (next >>> 7), next | 61);
+    return ((next ^ (next >>> 14)) >>> 0) / 4294967296;
+  };
+}
+```
+
+### PixiJS Init
+
+```ts
+await this.app.init({
+  autoDensity: true,
+  autoStart: false,
+  backgroundAlpha: 0,
+  preference: 'webgl',
+  preserveDrawingBuffer: true,
+  powerPreference: 'high-performance',
+  resizeTo: this.root,
+  resolution: Math.min(window.devicePixelRatio || 1, 2)
+});
+```
+
+### Input Handler
+
+```ts
+if (moved >= this.swipeThresholdSq) {
+  this.callbacks.onSwipe(this.start, end);
+} else {
+  this.callbacks.onTap(end);
+}
+```
+
+### Posterizer
+
+```ts
+const loadedFrames = await Promise.all(frames.slice(-3).map((frame) => loadImage(frame.dataUrl)));
+context.fillText(`survived ${formatTime(stats.survivalMs)}  •  phase ${stats.phase}`, 52, footerY + 154);
+return canvas.toDataURL('image/png', 0.86);
+```
+
+### GAS `doPost`
+
+```js
+function doPost(e) {
+  var payload = JSON.parse(e.postData.contents || '{}');
+  return ContentService.createTextOutput(JSON.stringify({
+    ok: true,
+    acceptedAt: new Date().toISOString(),
+    score: Number(payload.score || 0),
+    survivalMs: Number(payload.survivalMs || 0)
+  })).setMimeType(ContentService.MimeType.JSON);
+}
+```
+
+## Sample Replay Payload
+
+```json
+{
+  "version": 1,
+  "seed": "eh-2026-05-29-alpha",
+  "startedAt": 1780051200000,
+  "survivalMs": 67421,
+  "score": 1280,
+  "energyCaptured": 87,
+  "maxStreak": 24,
+  "tapEvents": [],
+  "swipeEvents": [],
+  "phaseTransitions": []
````

```diff
+}
+```
+
+## Test Results
+
+- `npm run build`: passed. TypeScript compiled and Vite produced `dist/`.
+- `npm run lint`: passed.
+- `npm run test`: passed, 2 files and 5 tests.
+- `npm run test:e2e`: passed, 4 mobile Chrome tests.
+- `npm run score:test`: passed, score endpoint returned status `200` and `{ "ok": true }`.
+- Local Chrome smoke test: passed via Playwright and in-app browser visual inspection.
+- Deterministic replay sanity: passed. Same seed plus same input timings produced identical replay payloads.
+- Touch simulation sanity: passed. Tap and swipe events are captured in replay payloads.
+- Posterizer export: passed. Browser test returns a `data:image/png;base64,...` poster.
+- Score submit test: passed against `/.netlify/functions/score-submit` handler import.
+
+## Performance Notes
+
+- The simulation uses a fixed 60 Hz step and avoids allocating per-frame input or entity arrays.
+- Orb and flyby sprites use generated textures rather than drawing new art every frame.
+- Background art is drawn once; per-frame `Graphics` clearing is limited to tethers, arms, HUD fill, and flyby trails.
+- No heavy filters or physics engine are used.
+- `preserveDrawingBuffer` supports poster screenshots but can cost some GPU performance; if poster export changes later, replacing it with targeted render-texture capture may be better.
+- The main visible bottleneck risk is `canvas.toDataURL()` during poster capture; it is user-triggered rather than hot-loop work.
+
+## Deployment Notes
+
+### Netlify
+
+- Build command: `VITE_BASE_PATH=/ npm run build`
+- Publish directory: `dist`
+- Functions directory: `netlify/functions`
+- Score function route: `/.netlify/functions/score-submit`
+- Branch deploys: connect the GitHub repo in Netlify and enable branch deploys for PR previews.
+- Local Netlify dev: `npx netlify dev` from the repo root.
+
+### GitHub Pages
+
+- Default Vite `base` is `/event-horizon/`.
+- Static build command: `npm run build`
+- Publish the generated `dist/` content through the chosen GitHub Pages workflow.
+- Environment assumption: GitHub Pages uses a subpath, while Netlify uses root. Override with `VITE_BASE_PATH=/` for root deploys.
+
+## PR Summary
+
+This PR introduces the first playable Event Horizon slice: deterministic simulation, PixiJS rendering, one-thumb input, replay payloads, phase-based collapse pressure, score submission samples, tests, documentation, screenshots, and this PDF report.
diff --git a/docs/pr-body.md b/docs/pr-body.md
new file mode 100644
index 0000000..8f6269d
--- /dev/null
+++ b/docs/pr-body.md
@@ -0,0 +1,19 @@
+## Summary
+
+- Adds a plain Vite + PixiJS v8 vertical slice for Event Horizon.
+- Implements deterministic 60 Hz simulation, seeded RNG, tap/swipe input capture, replay payloads, orb/flyby/shadow-arm gameplay, HUD, collapse animation, and poster export.
+- Adds Netlify score-submit function, Google Apps Script fallback sample, README, AGENTS.md, tests, screenshots, and a PDF developer report.
+
+## Tests
+
+- `npm run build`
+- `npm run lint`
+- `npm run test`
+- `npm run test:e2e`
+- `npm run score:test`
+
+## Notes
+
+- Default Vite base path supports GitHub Pages at `/event-horizon/`.
+- Netlify uses `VITE_BASE_PATH=/` through `netlify.toml`.
+- Score persistence is intentionally deferred; the current endpoint validates and acknowledges replay payloads.
diff --git a/eslint.config.js b/eslint.config.js
new file mode 100644
index 0000000..221d74e
--- /dev/null
+++ b/eslint.config.js
@@ -0,0 +1,37 @@
+import js from '@eslint/js';
+import tseslint from 'typescript-eslint';
+
+export default tseslint.config(
+  js.configs.recommended,
+  ...tseslint.configs.recommended,
+  {
+    ignores: ['dist/**', 'coverage/**', 'docs/**', 'playwright-report/**', 'test-results/**']
+  },
+  {
+    files: ['**/*.ts'],
```

```
+    languageOptions: {
+      parserOptions: {
+        projectService: true,
+        tsconfigRootDir: import.meta.dirname
+      }
+    },
+    rules: {
+      '@typescript-eslint/consistent-type-imports': 'error',
+      '@typescript-eslint/no-explicit-any': 'off'
+    }
+  },
+  {
+    files: ['netlify/functions/**/*.mjs', 'scripts/**/*.mjs'],
+    languageOptions: {
+      globals: {
+        console: 'readonly',
+        process: 'readonly',
+        Buffer: 'readonly',
+        Request: 'readonly',
+        Response: 'readonly',
+        URL: 'readonly',
+        window: 'readonly'
+      }
+    }
+  }
+);
diff --git a/gas/score-submit.gs b/gas/score-submit.gs
new file mode 100644
index 0000000..a4097a4
--- /dev/null
+++ b/gas/score-submit.gs
@@ -0,0 +1,29 @@
+/**
+ * Minimal Google Apps Script alternative score endpoint.
+ * Deploy as a Web App and set access to "Anyone" for anonymous submissions.
+ */
+function doPost(e) {
+  var payload = {};
+  try {
+    payload = JSON.parse(e.postData.contents || '{}');
+  } catch (error) {
+    return json_({ ok: false, error: 'invalid_json' });
+  }
+
+  if (payload.version !== 1 || typeof payload.seed !== 'string') {
+    return json_({ ok: false, error: 'invalid_replay' });
+  }
+
+  return json_({
+    ok: true,
+    acceptedAt: new Date().toISOString(),
+    score: Number(payload.score || 0),
+    survivalMs: Number(payload.survivalMs || 0)
+  });
+}
+
+function json_(body) {
+  return ContentService.createTextOutput(JSON.stringify(body)).setMimeType(
+    ContentService.MimeType.JSON
+  );
+}
diff --git a/index.html b/index.html
new file mode 100644
index 0000000..1ebe632
--- /dev/null
+++ b/index.html
@@ -0,0 +1,23 @@
+<!doctype html>
+<html lang="en-US">
+  <head>
+    <meta charset="UTF-8" />
+    <meta name="viewport" content="width=device-width, initial-scale=1, viewport-fit=cover" />
+    <meta name="theme-color" content="#03040a" />
+    <meta name="description" content="Event Horizon, a one-thumb galaxy survival prototype." />
+    <title>Event Horizon</title>
+  </head>
+  <body>
+    <main id="app" aria-label="Event Horizon game">
+      <div id="game-shell">
+        <div id="game-root" aria-label="Playable canvas"></div>
+        <div id="status-layer" aria-live="polite">
+          <button id="restart-button" type="button" aria-label="Restart run" title="Restart run">!»</button>
+          <button id="share-button" type="button" aria-label="Create share poster" title="Create share poster">!ê</button>
+          <a id="poster-link" download="event-horizon-poster.png" aria-label="Download share poster">!é</a>
+        </div>
+      </div>
+    </main>
+    <script type="module" src="/src/main.ts"></script>
+  </body>
+</html>
diff --git a/netlify.toml b/netlify.toml
new file mode 100644
```

```
index 0000000..538dee9
--- /dev/null
+++ b/netlify.toml
@@ -0,0 +1,12 @@
+[build]
+  command = "VITE_BASE_PATH=/ npm run build"
+  publish = "dist"
+
+[functions]
+  directory = "netlify/functions"
+
+[[headers]]
+  for = "/*"
+  [headers.values]
+    X-Content-Type-Options = "nosniff"
+    Referrer-Policy = "strict-origin-when-cross-origin"
diff --git a/netlify/functions/score-submit.mjs b/netlify/functions/score-submit.mjs
new file mode 100644
index 0000000..982d23f
--- /dev/null
+++ b/netlify/functions/score-submit.mjs
@@ -0,0 +1,67 @@
+const corsHeaders = {
+  'access-control-allow-origin': '*',
+  'access-control-allow-methods': 'POST, OPTIONS',
+  'access-control-allow-headers': 'content-type',
+  'content-type': 'application/json'
+};
+
+export default async function scoreSubmit(request) {
+  if (request.method === 'OPTIONS') {
+    return json({ ok: true }, 204);
+  }
+
+  if (request.method !== 'POST') {
+    return json({ ok: false, error: 'method_not_allowed' }, 405);
+  }
+
+  let payload;
+  try {
+    payload = await request.json();
+  } catch {
+    return json({ ok: false, error: 'invalid_json' }, 400);
+  }
+
+  const validationError = validateReplay(payload);
+  if (validationError) {
+    return json({ ok: false, error: validationError }, 422);
+  }
+
+  return json({
+    ok: true,
+    acceptedAt: new Date().toISOString(),
+    score: payload.score,
+    survivalMs: payload.survivalMs
+  });
+}
+
+function json(body, status = 200) {
+  return new Response(JSON.stringify(body), {
+    status,
+    headers: corsHeaders
+  });
+}
+
+function validateReplay(payload) {
+  if (!payload || typeof payload !== 'object') {
+    return 'payload_required';
+  }
+  if (payload.version !== 1) {
+    return 'version_required';
+  }
+  if (typeof payload.seed !== 'string' || payload.seed.length < 3) {
+    return 'seed_required';
+  }
+  if (!Number.isFinite(payload.startedAt)) {
+    return 'started_at_required';
+  }
+  if (!Number.isFinite(payload.survivalMs) || payload.survivalMs < 0) {
+    return 'survival_required';
+  }
+  if (!Number.isFinite(payload.score) || payload.score < 0) {
+    return 'score_required';
+  }
+  if (!Array.isArray(payload.tapEvents) || !Array.isArray(payload.swipeEvents)) {
+    return 'inputs_required';
+  }
+  return '';
+}
diff --git a/package.json b/package.json
new file mode 100644
index 0000000..949c0d9
```

```
--- /dev/null
+++ b/package.json
@@ -0,0 +1,30 @@
+{
+  "name": "event-horizon",
+  "version": "0.1.0",
+  "private": true,
+  "type": "module",
+  "scripts": {
+    "dev": "vite --host 127.0.0.1",
+    "build": "tsc --noEmit && vite build",
+    "preview": "vite preview --host 127.0.0.1",
+    "test": "vitest run",
+    "test:e2e": "playwright test",
+    "lint": "eslint .",
+    "score:test": "node scripts/test-score-submit.mjs",
+    "capture:artifacts": "node scripts/capture-artifacts.mjs",
+    "report:pdf": "node scripts/generate-report-pdf.mjs"
+  },
+  "dependencies": {
+    "pixi.js": "8.18.1"
+  },
+  "devDependencies": {
+    "@eslint/js": "9.39.1",
+    "@playwright/test": "1.57.0",
+    "eslint": "9.39.1",
+    "pdfkit": "0.17.2",
+    "typescript": "5.9.3",
+    "typescript-eslint": "8.49.0",
+    "vite": "8.0.14",
+    "vitest": "4.0.15"
+  }
+}
diff --git a/playwright.config.ts b/playwright.config.ts
new file mode 100644
index 0000000..148f315
--- /dev/null
+++ b/playwright.config.ts
@@ -0,0 +1,31 @@
+import { defineConfig, devices } from '@playwright/test';
+
+export default defineConfig({
+  testDir: './tests/e2e',
+  timeout: 30000,
+  fullyParallel: false,
+  use: {
+    baseURL: 'http://127.0.0.1:5173/event-horizon/',
+    channel: 'chrome',
+    viewport: { width: 390, height: 844 },
+    deviceScaleFactor: 2,
+    hasTouch: true,
+    isMobile: true,
+    trace: 'retain-on-failure'
+  },
+  projects: [
+    {
+      name: 'mobile-chrome',
+      use: {
+        ...devices['Pixel 7'],
+        channel: 'chrome'
+      }
+    }
+  ],
+  webServer: {
+    command: 'npm run dev -- --port 5173',
+    url: 'http://127.0.0.1:5173/event-horizon/',
+    reuseExistingServer: true,
+    timeout: 120000
+  }
+});
diff --git a/scripts/capture-artifacts.mjs b/scripts/capture-artifacts.mjs
new file mode 100644
index 0000000..821a5a0
--- /dev/null
+++ b/scripts/capture-artifacts.mjs
@@ -0,0 +1,55 @@
+import { mkdir, writeFile } from 'node:fs/promises';
+import { chromium } from '@playwright/test';
+
+const outputDir = new URL('../docs/artifacts/', import.meta.url);
+await mkdir(outputDir, { recursive: true });
+
+const browser = await chromium.launch({ channel: 'chrome' });
+const page = await browser.newPage({
+  viewport: { width: 390, height: 844 },
+  deviceScaleFactor: 2,
+  isMobile: true,
+  hasTouch: true
+});
+
+await page.goto('http://127.0.0.1:5173/event-horizon/', { waitUntil: 'networkidle' });
+await page.locator('canvas').waitFor({ state: 'visible', timeout: 10000 });
```

```
+await page.waitForTimeout(1300);
+await page.screenshot({
+  path: new URL('gameplay-mobile.jpg', outputDir).pathname,
+  type: 'jpeg',
+  quality: 58,
+  fullPage: false
+});
+
+const canvasBox = await page.locator('canvas').boundingBox();
+if (canvasBox) {
+  await page.touchscreen.tap(canvasBox.x + canvasBox.width * 0.48, canvasBox.y + canvasBox.height * 0.42);
+  await page.mouse.move(canvasBox.x + canvasBox.width * 0.28, canvasBox.y + canvasBox.height * 0.72);
+  await page.mouse.down();
+  await page.mouse.move(canvasBox.x + canvasBox.width * 0.76, canvasBox.y + canvasBox.height * 0.46, {
+    steps: 8
+  });
+  await page.mouse.up();
+}
+
+await page.waitForTimeout(900);
+const posterDataUrl = await page.evaluate(() => window.__EVENT_HORIZON__?.exportPoster());
+if (!posterDataUrl || !posterDataUrl.startsWith('data:image/png;base64,')) {
+  throw new Error('Poster export failed.');
+}
+await writeFile(new URL('share-poster.png', outputDir), Buffer.from(posterDataUrl.split(',')[1], 'base64'));
+
+await page.evaluate(() => window.__EVENT_HORIZON__?.forceEnd());
+await page.waitForTimeout(950);
+await page.screenshot({
+  path: new URL('collapse-mobile.jpg', outputDir).pathname,
+  type: 'jpeg',
+  quality: 56,
+  fullPage: false
+});
+
+await browser.close();
+console.log('Captured docs/artifacts/gameplay-mobile.jpg');
+console.log('Captured docs/artifacts/share-poster.png');
+console.log('Captured docs/artifacts/collapse-mobile.jpg');
diff --git a/scripts/generate-report-pdf.mjs b/scripts/generate-report-pdf.mjs
new file mode 100644
index 0000000..5cab74b
--- /dev/null
+++ b/scripts/generate-report-pdf.mjs
@@ -0,0 +1,180 @@
+import { execFileSync } from 'node:child_process';
+import { existsSync } from 'node:fs';
+import { mkdir, readFile, stat, writeFile } from 'node:fs/promises';
+import PDFDocument from 'pdfkit';
+
+const repoRoot = new URL('../', import.meta.url);
+const docsDir = new URL('../docs/', import.meta.url);
+const reportPath = new URL('first-pr-report.md', docsDir);
+const pdfPath = new URL('first-pr-report.pdf', docsDir);
+const artifacts = [
+  new URL('artifacts/gameplay-mobile.jpg', docsDir),
+  new URL('artifacts/share-poster.png', docsDir),
+  new URL('artifacts/collapse-mobile.jpg', docsDir)
+];
+
+await mkdir(docsDir, { recursive: true });
+
+const markdown = await readFile(reportPath, 'utf8');
+const diffStat = runGit(['diff', 'HEAD', '--stat', '--', '.', ':(exclude)docs/first-pr-report.pdf']);
+const diffTextRaw = runGit([
+  'diff',
+  'HEAD',
+  '--',
+  '.',
+  ':(exclude)package-lock.json',
+  ':(exclude)docs/artifacts/*.jpg',
+  ':(exclude)docs/artifacts/*.png',
+  ':(exclude)docs/first-pr-report.pdf'
+]);
+const maxDiffChars = 70000;
+const diffText =
+  diffTextRaw.length > maxDiffChars
+    ? `${diffTextRaw.slice(0, maxDiffChars)}\n\n[Diff truncated for PDF practicality. package-lock and binary images are
represented in the stat above.]\n`
+    : diffTextRaw;
+
+const doc = new PDFDocument({
+  autoFirstPage: false,
+  bufferPages: true,
+  compress: true,
+  margins: { top: 48, right: 48, bottom: 54, left: 48 },
+  size: 'LETTER'
+});
+
+const chunks = [];
+doc.on('data', (chunk) => chunks.push(chunk));
+
```

```
+addPage(doc);
+doc.font('Helvetica-Bold').fontSize(20).text('Event Horizon First PR Report');
+doc.moveDown(0.4);
+doc.font('Helvetica').fontSize(9).fillColor('#444').text(`Generated: ${new Date().toISOString()}`);
+doc.moveDown(1);
+
+writeMarkdown(doc, markdown);
+addScreenshots(doc);
+addDiffSection(doc, diffStat, diffText);
+addPageNumbers(doc);
+
+doc.end();
+const pdfBuffer = await new Promise((resolve) => {
+  doc.on('end', () => resolve(Buffer.concat(chunks)));
+});
+await writeFile(pdfPath, pdfBuffer);
+const size = await stat(pdfPath);
+console.log(`Wrote docs/first-pr-report.pdf (${Math.round(size.size / 1024)} KiB)`);
+
+function runGit(args) {
+  try {
+    return execFileSync('git', args, {
+      cwd: repoRoot,
+      encoding: 'utf8',
+      maxBuffer: 1024 * 1024 * 8
+    });
+  } catch (error) {
+    return `Unable to collect git output: ${error.message}\n`;
+  }
+}
+
+function addPage(document) {
+  document.addPage();
+  document.fillColor('#111');
+}
+
+function writeMarkdown(document, source) {
+  const lines = source.split('\n');
+  let inCode = false;
+  for (const line of lines) {
+    if (line.startsWith('```')) {
+      inCode = !inCode;
+      document.moveDown(0.15);
+      continue;
+    }
+
+    if (line.startsWith('# ')) {
+      ensureRoom(document, 44);
+      document.moveDown(0.6);
+      document.font('Helvetica-Bold').fontSize(16).fillColor('#111').text(line.replace(/^# /, ''));
+      document.moveDown(0.3);
+    } else if (line.startsWith('## ')) {
+      ensureRoom(document, 38);
+      document.moveDown(0.4);
+      document.font('Helvetica-Bold').fontSize(13).fillColor('#111').text(line.replace(/^## /, ''));
+      document.moveDown(0.2);
+    } else if (line.startsWith('### ')) {
+      ensureRoom(document, 32);
+      document.font('Helvetica-Bold').fontSize(11).fillColor('#111').text(line.replace(/^### /, ''));
+    } else if (inCode) {
+      ensureRoom(document, 18);
+      document.font('Courier').fontSize(7.2).fillColor('#222').text(line || ' ', {
+        lineGap: 1
+      });
+    } else if (line.trim().length === 0) {
+      document.moveDown(0.35);
+    } else {
+      ensureRoom(document, 20);
+      document.font(line.startsWith('- ') || /^\d+\./.test(line) ? 'Helvetica' :
'Helvetica').fontSize(8.7).fillColor('#222').text(line, {
+        lineGap: 2
+      });
+    }
+  }
+}
+
+function addScreenshots(document) {
+  addPage(document);
+  document.font('Helvetica-Bold').fontSize(16).fillColor('#111').text('Screenshots and Poster');
+  document.moveDown(0.5);
+  const labels = ['Gameplay mobile screenshot', 'Three-frame share poster', 'Collapse mobile screenshot'];
+  for (let index = 0; index < artifacts.length; index += 1) {
+    const image = artifacts[index];
+    if (!existsSync(image)) {
+      continue;
+    }
+    ensureRoom(document, 250);
+    document.font('Helvetica-Bold').fontSize(10).text(labels[index]);
+    document.moveDown(0.2);
+    document.image(image.pathname, {
+      fit: index === 1 ? [220, 300] : [180, 300],
+      align: 'center'
```

```
+    });
+    document.moveDown(0.7);
+  }
+}
+
+function addDiffSection(document, statText, fullDiff) {
+  addPage(document);
+  document.font('Helvetica-Bold').fontSize(16).fillColor('#111').text('File Diffs');
+  document.moveDown(0.5);
+  document.font('Helvetica-Bold').fontSize(11).text('Diff Stat');
+  document.moveDown(0.2);
+  writePre(document, statText || 'No diff stat available.');
+  document.moveDown(0.7);
+  document.font('Helvetica-Bold').fontSize(11).text('Unified Diff');
+  document.moveDown(0.2);
+  writePre(document, fullDiff || 'No text diff available.');
+}
+
+function writePre(document, text) {
+  for (const line of text.split('\n')) {
+    ensureRoom(document, 14);
+    document.font('Courier').fontSize(6.5).fillColor('#222').text(line || ' ', {
+      lineGap: 0.5
+    });
+  }
+}
+
+function ensureRoom(document, height) {
+  if (document.y + height > document.page.height - document.page.margins.bottom) {
+    addPage(document);
+  }
+}
+
+function addPageNumbers(document) {
+  const range = document.bufferedPageRange();
+  for (let index = range.start; index < range.start + range.count; index += 1) {
+    document.switchToPage(index);
+    document.font('Helvetica').fontSize(8).fillColor('#666').text(`Event Horizon first PR report  •  ${index + 1}`, 48, 744, {
+      align: 'center',
+      width: 516
+    });
+  }
+}
diff --git a/scripts/test-score-submit.mjs b/scripts/test-score-submit.mjs
new file mode 100644
index 0000000..4ba1ba0
--- /dev/null
+++ b/scripts/test-score-submit.mjs
@@ -0,0 +1,30 @@
+import scoreSubmit from '../netlify/functions/score-submit.mjs';
+
+const replay = {
+  version: 1,
+  seed: 'eh-2026-05-29-alpha',
+  startedAt: 1780051200000,
+  survivalMs: 67421,
+  score: 1280,
+  energyCaptured: 87,
+  maxStreak: 24,
+  tapEvents: [],
+  swipeEvents: [],
+  phaseTransitions: []
+};
+
+const response = await scoreSubmit(
+  new Request('https://event-horizon.test/.netlify/functions/score-submit', {
+    method: 'POST',
+    headers: { 'content-type': 'application/json' },
+    body: JSON.stringify(replay)
+  })
+);
+
+const body = await response.json();
+if (response.status !== 200 || !body.ok || body.score !== replay.score) {
+  console.error(JSON.stringify({ status: response.status, body }, null, 2));
+  process.exit(1);
+}
+
+console.log(JSON.stringify({ status: response.status, body }, null, 2));
diff --git a/src/game/EventHorizonGame.ts b/src/game/EventHorizonGame.ts
new file mode 100644
index 0000000..61fdc93
--- /dev/null
+++ b/src/game/EventHorizonGame.ts
@@ -0,0 +1,475 @@
+import {
+  Application,
+  Container,
+  Graphics,
+  Sprite,
+  Text,
+  TextStyle,
```

```
+  type Texture,
+  type Renderer
+} from 'pixi.js';
+import {
+  BLACK_HOLE_X,
+  BLACK_HOLE_Y,
+  MAX_ENERGY,
+  WORLD_HEIGHT,
+  WORLD_WIDTH
+} from './constants';
+import { FixedStepLoop } from './FixedStepLoop';
+import { InputHandler, type WorldPoint } from './InputHandler';
+import { clamp, formatTime } from './math';
+import { createSharePoster, type PosterFrame } from './posterizer';
+import { submitScore } from './scoreClient';
+import { Simulation, type SimulationOptions } from './Simulation';
+import type { FlybyState, OrbState, ReplayPayload, ShadowArmState, SimulationSnapshot } from './types';
+
+interface OrbView {
+  sprite: Sprite;
+  glow: Sprite;
+}
+
+interface FlybyView {
+  sprite: Sprite;
+  streak: Graphics;
+}
+
+export class EventHorizonGame {
+  private readonly app = new Application<Renderer>();
+  private readonly world = new Container();
+  private readonly background = new Graphics();
+  private readonly tetherLayer = new Graphics();
+  private readonly armLayer = new Graphics();
+  private readonly blackHole = new Graphics();
+  private readonly collapseLayer = new Graphics();
+  private readonly hud = new Container();
+  private readonly energyBarBack = new Graphics();
+  private readonly energyBarFill = new Graphics();
+  private readonly scoreText = new Text({
+    text: '0',
+    style: new TextStyle({
+      fill: '#f7fbff',
+      fontFamily: 'Inter, system-ui, sans-serif',
+      fontSize: 52,
+      fontWeight: '800'
+    })
+  });
+  private readonly metaText = new Text({
+    text: '',
+    style: new TextStyle({
+      fill: '#9fe7ff',
+      fontFamily: 'Inter, system-ui, sans-serif',
+      fontSize: 26,
+      fontWeight: '600'
+    })
+  });
+  private readonly endText = new Text({
+    text: '',
+    style: new TextStyle({
+      align: 'center',
+      fill: '#f7fbff',
+      fontFamily: 'Inter, system-ui, sans-serif',
+      fontSize: 54,
+      fontWeight: '800'
+    })
+  });
+  private readonly sim: Simulation;
+  private readonly loop: FixedStepLoop;
+  private input?: InputHandler;
+  private orbViews: OrbView[] = [];
+  private flybyViews: FlybyView[] = [];
+  private frameSamples: PosterFrame[] = [];
+  private sampleCooldownMs = 0;
+  private scale = 1;
+  private offsetX = 0;
+  private offsetY = 0;
+  private scoreSubmitted = false;
+
+  constructor(
+    private readonly root: HTMLElement,
+    options: SimulationOptions
+  ) {
+    this.sim = new Simulation(options);
+    this.loop = new FixedStepLoop(
+      (dtMs) => this.step(dtMs),
+      () => this.render()
+    );
+  }
+
+  async start(): Promise<void> {
+    await this.app.init({
```

```
+      autoDensity: true,
+      autoStart: false,
+      backgroundAlpha: 0,
+      clearBeforeRender: true,
+      hello: false,
+      preference: 'webgl',
+      preserveDrawingBuffer: true,
+      powerPreference: 'high-performance',
+      resizeTo: this.root,
+      resolution: Math.min(window.devicePixelRatio || 1, 2)
+    });
+
+    this.root.appendChild(this.app.canvas);
+    this.buildScene();
+    this.input = new InputHandler(this.app.canvas, {
+      screenToWorld: (clientX, clientY) => this.screenToWorld(clientX, clientY),
+      onTap: (point) => this.sim.applyTap(point.x, point.y),
+      onSwipe: (start, end) => this.sim.applySwipe(start.x, start.y, end.x, end.y)
+    });
+    window.addEventListener('resize', this.resize);
+    this.resize();
+    this.loop.start();
+  }
+
+  restart(): void {
+    this.sim.reset();
+    this.scoreSubmitted = false;
+    this.frameSamples = [];
+    this.sampleCooldownMs = 0;
+    this.loop.resetClock();
+  }
+
+  forceEnd(): void {
+    this.sim.forceEnd();
+  }
+
+  destroy(): void {
+    this.loop.stop();
+    this.input?.destroy();
+    window.removeEventListener('resize', this.resize);
+    this.app.destroy(true, { children: true, texture: true });
+  }
+
+  getReplayPayload(): ReplayPayload {
+    return this.sim.getReplayPayload();
+  }
+
+  getSnapshot(): SimulationSnapshot {
+    return this.sim.getSnapshot();
+  }
+
+  async exportPoster(): Promise<string> {
+    this.sampleFrame('current');
+    const snapshot = this.sim.getSnapshot();
+    const frames = this.frameSamples.length >= 3 ? this.frameSamples : this.makeFallbackFrames();
+    return createSharePoster(frames, {
+      score: snapshot.score,
+      survivalMs: this.sim.getReplayPayload().survivalMs,
+      seed: this.sim.seed,
+      phase: snapshot.phase
+    });
+  }
+
+  private buildScene(): void {
+    this.world.sortableChildren = false;
+    this.app.stage.addChild(this.world);
+    this.world.addChild(this.background);
+    this.world.addChild(this.armLayer);
+    this.world.addChild(this.blackHole);
+    this.world.addChild(this.tetherLayer);
+    this.createOrbViews();
+    this.createFlybyViews();
+    this.world.addChild(this.hud);
+    this.hud.addChild(this.energyBarBack, this.energyBarFill, this.scoreText, this.metaText, this.endText);
+    this.world.addChild(this.collapseLayer);
+    this.drawBackground();
+    this.drawHudBack();
+    this.endText.anchor.set(0.5);
+    this.endText.position.set(WORLD_WIDTH / 2, WORLD_HEIGHT * 0.58);
+  }
+
+  private createOrbViews(): void {
+    const orbTexture = this.makeOrbTexture(0x78f2ff, 32, false);
+    const glowTexture = this.makeOrbTexture(0xbaf8ff, 76, true);
+    for (let index = 0; index < 24; index += 1) {
+      const glow = new Sprite(glowTexture);
+      glow.anchor.set(0.5);
+      glow.blendMode = 'add';
+      glow.visible = false;
+      const sprite = new Sprite(orbTexture);
+      sprite.anchor.set(0.5);
+      sprite.visible = false;
```

```
+      this.world.addChild(glow, sprite);
+      this.orbViews.push({ sprite, glow });
+    }
+  }
+
+  private createFlybyViews(): void {
+    const texture = this.makeOrbTexture(0xfff0a3, 24, false);
+    for (let index = 0; index < 6; index += 1) {
+      const streak = new Graphics();
+      const sprite = new Sprite(texture);
+      sprite.anchor.set(0.5);
+      streak.visible = false;
+      sprite.visible = false;
+      this.world.addChild(streak, sprite);
+      this.flybyViews.push({ sprite, streak });
+    }
+  }
+
+  private makeOrbTexture(color: number, radius: number, soft: boolean): Texture {
+    const graphics = new Graphics();
+    const alpha = soft ? 0.14 : 0.9;
+    graphics.circle(radius, radius, radius).fill({ color, alpha });
+    graphics.circle(radius, radius, radius * 0.48).fill({ color: 0xffffff, alpha: soft ? 0.12 : 0.65 });
+    return this.app.renderer.generateTexture(graphics);
+  }
+
+  private drawBackground(): void {
+    this.background.clear();
+    this.background.rect(0, 0, WORLD_WIDTH, WORLD_HEIGHT).fill(0x03040a);
+
+    for (let index = 0; index < 240; index += 1) {
+      const x = (index * 197.63) % WORLD_WIDTH;
+      const y = (index * 311.19) % WORLD_HEIGHT;
+      const depth = (index % 11) / 10;
+      const alpha = 0.18 + depth * 0.38;
+      const size = 1.1 + depth * 2.2;
+      this.background.circle(x, y, size).fill({ color: 0xbfeaff, alpha });
+    }
+
+    for (let index = 0; index < 150; index += 1) {
+      const t = index / 149;
+      const angle = t * Math.PI * 8.8;
+      const radius = 82 + t * 690;
+      const x = BLACK_HOLE_X + Math.cos(angle) * radius;
+      const y = BLACK_HOLE_Y + Math.sin(angle) * radius * 0.57;
+      const color = index % 2 === 0 ? 0x305da6 : 0x6e366f;
+      this.background.circle(x, y, 3 + t * 7).fill({ color, alpha: 0.07 + (1 - t) * 0.08 });
+      this.background.circle(WORLD_WIDTH - x, WORLD_HEIGHT * 0.88 - y * 0.34, 2 + t * 4).fill({
+        color: 0x1aaf96,
+        alpha: 0.035
+      });
+    }
+  }
+
+  private drawHudBack(): void {
+    this.energyBarBack.clear();
+    this.energyBarBack.roundRect(78, WORLD_HEIGHT - 132, WORLD_WIDTH - 156, 38, 7).fill({
+      color: 0x061120,
+      alpha: 0.86
+    });
+    this.energyBarBack.roundRect(78, WORLD_HEIGHT - 132, WORLD_WIDTH - 156, 38, 7).stroke({
+      color: 0x78f2ff,
+      alpha: 0.36,
+      width: 2
+    });
+    this.scoreText.position.set(72, 70);
+    this.metaText.position.set(76, 132);
+  }
+
+  private step(dtMs: number): void {
+    const wasEnded = this.sim.getSnapshot().ended;
+    this.sim.step(dtMs);
+    const snapshot = this.sim.getSnapshot();
+    this.sampleCooldownMs -= dtMs;
+    if (this.sampleCooldownMs <= 0 && !snapshot.ended) {
+      this.sampleFrame(`t${Math.round(snapshot.timeMs)}`);
+      this.sampleCooldownMs = 1400;
+    }
+    if (!wasEnded && snapshot.ended && !this.scoreSubmitted) {
+      this.scoreSubmitted = true;
+      void submitScore(this.sim.getReplayPayload());
+    }
+  }
+
+  private render(): void {
+    const snapshot = this.sim.getSnapshot();
+    this.world.position.set(this.offsetX, this.offsetY);
+    this.world.scale.set(this.scale);
+    this.renderShadowArms(snapshot.shadowArms, snapshot.phase);
+    this.renderBlackHole(snapshot);
+    this.renderTethers(snapshot.orbs);
+    this.renderOrbs(snapshot.orbs, snapshot.collapseT);
```

```
+    this.renderFlybys(snapshot.flybys);
+    this.renderHud(snapshot);
+    this.renderCollapse(snapshot);
+    this.app.render();
+  }
+
+  private renderShadowArms(arms: readonly ShadowArmState[], phase: number): void {
+    this.armLayer.clear();
+    if (phase < 2) {
+      return;
+    }
+    for (const arm of arms) {
+      const length = 420 + phase * 90;
+      const endX = BLACK_HOLE_X + Math.cos(arm.angle) * length;
+      const endY = BLACK_HOLE_Y + Math.sin(arm.angle) * length;
+      const alpha = arm.stunMs > 0 ? 0.12 : 0.12 + arm.intensity * 0.32;
+      this.armLayer.moveTo(BLACK_HOLE_X, BLACK_HOLE_Y);
+      this.armLayer.lineTo(endX, endY);
+      this.armLayer.stroke({ color: arm.stunMs > 0 ? 0x8ff9ff : 0x1b102b, alpha, width: 50 + phase * 7 });
+      this.armLayer.moveTo(BLACK_HOLE_X, BLACK_HOLE_Y);
+      this.armLayer.lineTo(endX, endY);
+      this.armLayer.stroke({ color: 0x8d62ff, alpha: alpha * 0.45, width: 8 });
+    }
+  }
+
+  private renderBlackHole(snapshot: SimulationSnapshot): void {
+    this.blackHole.clear();
+    const phaseT = (snapshot.phase - 1) / 3;
+    const pulse = Math.sin(snapshot.timeMs * 0.004) * 0.5 + 0.5;
+    const radius = 80 + phaseT * 54 + snapshot.collapseT * 260;
+    this.blackHole.circle(BLACK_HOLE_X, BLACK_HOLE_Y, radius * 2.2).fill({
+      color: 0x0a1424,
+      alpha: 0.08 + phaseT * 0.13
+    });
+    this.blackHole.circle(BLACK_HOLE_X, BLACK_HOLE_Y, radius * 1.25).stroke({
+      color: 0x6bcfff,
+      alpha: 0.14 + phaseT * 0.26,
+      width: 7 + pulse * 6
+    });
+    this.blackHole.circle(BLACK_HOLE_X, BLACK_HOLE_Y, radius).fill({ color: 0x000000, alpha: 0.82 });
+    this.blackHole.circle(BLACK_HOLE_X, BLACK_HOLE_Y, radius * 0.38).fill({ color: 0x03040a, alpha: 1 });
+  }
+
+  private renderTethers(orbs: readonly OrbState[]): void {
+    this.tetherLayer.clear();
+    for (const orb of orbs) {
+      if (!orb.active) {
+        continue;
+      }
+      const alpha = orb.captured ? 0.72 : orb.frozenMs > 0 ? 0.5 : 0.19;
+      this.tetherLayer.moveTo(BLACK_HOLE_X, BLACK_HOLE_Y);
+      this.tetherLayer.lineTo(orb.x, orb.y);
+      this.tetherLayer.stroke({
+        color: orb.captured ? 0xd9fbff : 0x5bc7ff,
+        alpha,
+        width: orb.captured ? 6 : orb.frozenMs > 0 ? 4 : 2
+      });
+    }
+  }
+
+  private renderOrbs(orbs: readonly OrbState[], collapseT: number): void {
+    for (let index = 0; index < this.orbViews.length; index += 1) {
+      const orb = orbs[index];
+      const view = this.orbViews[index];
+      if (!orb?.active) {
+        view.sprite.visible = false;
+        view.glow.visible = false;
+        continue;
+      }
+      const bendX = BLACK_HOLE_X + (orb.x - BLACK_HOLE_X) * (1 - collapseT * 0.88);
+      const bendY = BLACK_HOLE_Y + (orb.y - BLACK_HOLE_Y) * (1 - collapseT * 0.88);
+      const frozen = orb.frozenMs > 0;
+      view.sprite.visible = true;
+      view.glow.visible = true;
+      view.sprite.position.set(bendX, bendY);
+      view.glow.position.set(bendX, bendY);
+      const scale = orb.radius / 32;
+      view.sprite.scale.set(scale * (orb.captured ? 1.35 : frozen ? 1.22 : 1));
+      view.glow.scale.set(scale * (orb.captured ? 1.9 : frozen ? 1.45 : 1.08));
+      view.glow.alpha = orb.captured ? 1 : frozen ? 0.95 : 0.34;
+      view.sprite.alpha = 1 - collapseT * 0.45;
+    }
+  }
+
+  private renderFlybys(flybys: readonly FlybyState[]): void {
+    for (let index = 0; index < this.flybyViews.length; index += 1) {
+      const flyby = flybys[index];
+      const view = this.flybyViews[index];
+      if (!flyby?.active) {
+        view.sprite.visible = false;
+        view.streak.visible = false;
+        continue;
```

```
+      }
+      view.sprite.visible = true;
+      view.streak.visible = true;
+      view.sprite.position.set(flyby.x, flyby.y);
+      view.sprite.scale.set(flyby.radius / 24);
+      view.streak.clear();
+      view.streak.moveTo(flyby.x - flyby.vx * 0.08, flyby.y - flyby.vy * 0.08);
+      view.streak.lineTo(flyby.x, flyby.y);
+      view.streak.stroke({ color: 0xfff0a3, alpha: 0.55, width: 8 });
+    }
+  }
+
+  private renderHud(snapshot: SimulationSnapshot): void {
+    const width = (WORLD_WIDTH - 172) * clamp(snapshot.energy / MAX_ENERGY, 0, 1);
+    const barColor = snapshot.phase >= 4 ? 0xff5d73 : snapshot.phase >= 3 ? 0xffc857 : 0x67f4ff;
+    this.energyBarFill.clear();
+    this.energyBarFill.roundRect(86, WORLD_HEIGHT - 124, width, 22, 6).fill({ color: barColor, alpha: 0.95 });
+    this.energyBarFill.roundRect(86, WORLD_HEIGHT - 124, width, 22, 6).fill({ color: 0xffffff, alpha: 0.14 });
+    this.scoreText.text = snapshot.score.toString();
+    this.metaText.text = `${formatTime(this.sim.getReplayPayload().survivalMs)}  PHASE ${snapshot.phase}  x${snapshot.streak}`;
+    this.endText.visible = snapshot.ended;
+    this.endText.text = snapshot.ended
+      ? `GALAXY COLLAPSED\n${snapshot.score}  •  ${formatTime(this.sim.getReplayPayload().survivalMs)}`
+      : '';
+  }
+
+  private renderCollapse(snapshot: SimulationSnapshot): void {
+    this.collapseLayer.clear();
+    if (!snapshot.ended) {
+      return;
+    }
+    const t = snapshot.collapseT;
+    this.world.scale.set(this.scale * (1 - t * 0.06));
+    this.world.position.set(this.offsetX + WORLD_WIDTH * this.scale * t * 0.03, this.offsetY + WORLD_HEIGHT * this.scale * t *
0.025);
+    this.collapseLayer.rect(0, 0, WORLD_WIDTH, WORLD_HEIGHT).fill({ color: 0x000000, alpha: clamp((t - 0.42) / 0.58, 0, 1) });
+  }
+
+  private readonly resize = (): void => {
+    const rect = this.root.getBoundingClientRect();
+    this.scale = Math.min(rect.width / WORLD_WIDTH, rect.height / WORLD_HEIGHT);
+    const viewWidth = WORLD_WIDTH * this.scale;
+    const viewHeight = WORLD_HEIGHT * this.scale;
+    this.offsetX = (rect.width - viewWidth) / 2;
+    this.offsetY = (rect.height - viewHeight) / 2;
+    this.world.position.set(this.offsetX, this.offsetY);
+    this.world.scale.set(this.scale);
+  };
+
+  private screenToWorld(clientX: number, clientY: number): WorldPoint {
+    const rect = this.app.canvas.getBoundingClientRect();
+    return {
+      x: clamp((clientX - rect.left - this.offsetX) / this.scale, 0, WORLD_WIDTH),
+      y: clamp((clientY - rect.top - this.offsetY) / this.scale, 0, WORLD_HEIGHT)
+    };
+  }
+
+  private sampleFrame(label: string): void {
+    try {
+      const dataUrl = this.app.canvas.toDataURL('image/png', 0.72);
+      this.frameSamples.push({ dataUrl, label });
+      if (this.frameSamples.length > 3) {
+        this.frameSamples.shift();
+      }
+    } catch {
+      this.frameSamples = this.makeFallbackFrames();
+    }
+  }
+
+  private makeFallbackFrames(): PosterFrame[] {
+    const snapshot = this.sim.getSnapshot();
+    return [0, 1, 2].map((index) => ({
+      dataUrl: this.makeMockFrame(snapshot, index),
+      label: `mock-${index}`
+    }));
+  }
+
+  private makeMockFrame(snapshot: SimulationSnapshot, index: number): string {
+    const canvas = document.createElement('canvas');
+    canvas.width = 540;
+    canvas.height = 960;
+    const context = canvas.getContext('2d');
+    if (!context) {
+      return '';
+    }
+    context.fillStyle = '#03040a';
+    context.fillRect(0, 0, canvas.width, canvas.height);
+    context.fillStyle = index === 0 ? '#143c5c' : index === 1 ? '#3c285f' : '#501421';
+    context.beginPath();
+    context.arc(270, 410, 180 + snapshot.phase * 24, 0, Math.PI * 2);
+    context.fill();
+    context.fillStyle = '#000000';
```

```
+    context.beginPath();
+    context.arc(270, 410, 62 + snapshot.phase * 15, 0, Math.PI * 2);
+    context.fill();
+    context.fillStyle = '#f7fbff';
+    context.font = '700 54px system-ui';
+    context.fillText(String(snapshot.score), 52, 830);
+    return canvas.toDataURL('image/png', 0.8);
+  }
+}
diff --git a/src/game/FixedStepLoop.ts b/src/game/FixedStepLoop.ts
new file mode 100644
index 0000000..ab35246
--- /dev/null
+++ b/src/game/FixedStepLoop.ts
@@ -0,0 +1,54 @@
+import { FIXED_STEP_MS } from './constants';
+
+export class FixedStepLoop {
+  private accumulatorMs = 0;
+  private frameId = 0;
+  private lastNow = 0;
+  private readonly maxFrameMs = 250;
+  private running = false;
+
+  constructor(
+    private readonly step: (dtMs: number) => void,
+    private readonly render: (alpha: number) => void
+  ) {}
+
+  start(): void {
+    if (this.running) {
+      return;
+    }
+    this.running = true;
+    this.lastNow = performance.now();
+    this.frameId = requestAnimationFrame(this.tick);
+  }
+
+  stop(): void {
+    if (!this.running) {
+      return;
+    }
+    this.running = false;
+    cancelAnimationFrame(this.frameId);
+  }
+
+  resetClock(): void {
+    this.accumulatorMs = 0;
+    this.lastNow = performance.now();
+  }
+
+  private readonly tick = (now: number): void => {
+    if (!this.running) {
+      return;
+    }
+
+    const elapsed = Math.min(now - this.lastNow, this.maxFrameMs);
+    this.lastNow = now;
+    this.accumulatorMs += elapsed;
+
+    while (this.accumulatorMs >= FIXED_STEP_MS) {
+      this.step(FIXED_STEP_MS);
+      this.accumulatorMs -= FIXED_STEP_MS;
+    }
+
+    this.render(this.accumulatorMs / FIXED_STEP_MS);
+    this.frameId = requestAnimationFrame(this.tick);
+  };
+}
diff --git a/src/game/InputHandler.ts b/src/game/InputHandler.ts
new file mode 100644
index 0000000..5fe81f8
--- /dev/null
+++ b/src/game/InputHandler.ts
@@ -0,0 +1,79 @@
+import { distanceSquared } from './math';
+
+export interface WorldPoint {
+  x: number;
+  y: number;
+}
+
+export interface InputCallbacks {
+  screenToWorld: (clientX: number, clientY: number) => WorldPoint;
+  onTap: (point: WorldPoint) => void;
+  onSwipe: (start: WorldPoint, end: WorldPoint) => void;
+}
+
+export class InputHandler {
+  private activePointerId: number | null = null;
+  private start: WorldPoint | null = null;
+  private readonly swipeThresholdSq = 44 * 44;
```

```
+
+  constructor(
+    private readonly target: HTMLElement,
+    private readonly callbacks: InputCallbacks
+  ) {
+    this.target.addEventListener('pointerdown', this.handlePointerDown, { passive: false });
+    this.target.addEventListener('pointermove', this.handlePointerMove, { passive: false });
+    this.target.addEventListener('pointerup', this.handlePointerUp, { passive: false });
+    this.target.addEventListener('pointercancel', this.handlePointerCancel, { passive: false });
+  }
+
+  destroy(): void {
+    this.target.removeEventListener('pointerdown', this.handlePointerDown);
+    this.target.removeEventListener('pointermove', this.handlePointerMove);
+    this.target.removeEventListener('pointerup', this.handlePointerUp);
+    this.target.removeEventListener('pointercancel', this.handlePointerCancel);
+  }
+
+  private readonly handlePointerDown = (event: PointerEvent): void => {
+    event.preventDefault();
+    if (this.activePointerId !== null) {
+      return;
+    }
+    this.activePointerId = event.pointerId;
+    this.target.setPointerCapture?.(event.pointerId);
+    this.start = this.callbacks.screenToWorld(event.clientX, event.clientY);
+  };
+
+  private readonly handlePointerMove = (event: PointerEvent): void => {
+    if (event.pointerId !== this.activePointerId) {
+      return;
+    }
+    event.preventDefault();
+  };
+
+  private readonly handlePointerUp = (event: PointerEvent): void => {
+    if (event.pointerId !== this.activePointerId || !this.start) {
+      return;
+    }
+    event.preventDefault();
+    const end = this.callbacks.screenToWorld(event.clientX, event.clientY);
+    const moved = distanceSquared(this.start.x, this.start.y, end.x, end.y);
+    if (moved >= this.swipeThresholdSq) {
+      this.callbacks.onSwipe(this.start, end);
+    } else {
+      this.callbacks.onTap(end);
+    }
+    this.resetPointer(event.pointerId);
+  };
+
+  private readonly handlePointerCancel = (event: PointerEvent): void => {
+    if (event.pointerId === this.activePointerId) {
+      this.resetPointer(event.pointerId);
+    }
+  };
+
+  private resetPointer(pointerId: number): void {
+    this.target.releasePointerCapture?.(pointerId);
+    this.activePointerId = null;
+    this.start = null;
+  }
+}
diff --git a/src/game/Simulation.ts b/src/game/Simulation.ts
new file mode 100644
index 0000000..125f43b
--- /dev/null
+++ b/src/game/Simulation.ts
@@ -0,0 +1,553 @@
+import {
+  BLACK_HOLE_X,
+  BLACK_HOLE_Y,
+  FLYBY_POOL_SIZE,
+  INITIAL_ENERGY,
+  MAX_ENERGY,
+  ORB_POOL_SIZE,
+  WORLD_HEIGHT,
+  WORLD_WIDTH
+} from './constants';
+import { clamp, distanceSquared, distanceToSegmentSquared } from './math';
+import { createSeededRandom, type RandomSource } from './rng';
+import type {
+  FlybyState,
+  GamePhase,
+  OrbState,
+  PhaseTransition,
+  ReplayPayload,
+  ShadowArmState,
+  SimulationSnapshot,
+  SwipeEvent,
+  TapEvent
+} from './types';
+
```

```
+const ORB_CAPTURE_RADIUS = 82;
+const FLYBY_CAPTURE_RADIUS = 58;
+const EVENT_HORIZON_RADIUS = 98;
+const ARM_LENGTH = 760;
+const ARM_HIT_WIDTH = 46;
+
+export interface SimulationOptions {
+  seed: string;
+  startedAt: number;
+}
+
+export interface TimedInput {
+  kind: 'tap' | 'swipe';
+  t: number;
+  x: number;
+  y: number;
+  x2?: number;
+  y2?: number;
+}
+
+export class Simulation {
+  readonly seed: string;
+  readonly startedAt: number;
+  private rng: RandomSource;
+  private nextOrbId = 1;
+  private nextFlybyId = 1;
+  private orbSpawnMs = 900;
+  private flybySpawnMs = 2600;
+  private readonly tapEvents: TapEvent[] = [];
+  private readonly swipeEvents: SwipeEvent[] = [];
+  private readonly phaseTransitions: PhaseTransition[] = [];
+  private readonly orbs: OrbState[] = [];
+  private readonly flybys: FlybyState[] = [];
+  private readonly shadowArms: ShadowArmState[] = [];
+  private timeMs = 0;
+  private score = 0;
+  private energy = INITIAL_ENERGY;
+  private energyCaptured = 0;
+  private streak = 0;
+  private maxStreak = 0;
+  private phase: GamePhase = 1;
+  private ended = false;
+  private collapseMs = 0;
+
+  constructor(options: SimulationOptions) {
+    this.seed = options.seed;
+    this.startedAt = options.startedAt;
+    this.rng = createSeededRandom(options.seed);
+    this.initPools();
+    this.phaseTransitions.push({ t: 0, phase: 1, energy: this.energy });
+  }
+
+  reset(): void {
+    this.rng = createSeededRandom(this.seed);
+    this.nextOrbId = 1;
+    this.nextFlybyId = 1;
+    this.orbSpawnMs = 900;
+    this.flybySpawnMs = 2600;
+    this.tapEvents.length = 0;
+    this.swipeEvents.length = 0;
+    this.phaseTransitions.length = 0;
+    this.timeMs = 0;
+    this.score = 0;
+    this.energy = INITIAL_ENERGY;
+    this.energyCaptured = 0;
+    this.streak = 0;
+    this.maxStreak = 0;
+    this.phase = 1;
+    this.ended = false;
+    this.collapseMs = 0;
+    this.initPools();
+    this.phaseTransitions.push({ t: 0, phase: 1, energy: this.energy });
+  }
+
+  step(dtMs: number): void {
+    const dt = dtMs / 1000;
+    if (this.ended) {
+      this.timeMs += dtMs;
+      this.collapseMs = Math.min(this.collapseMs + dtMs, 2200);
+      return;
+    }
+
+    this.timeMs += dtMs;
+    this.energy = clamp(this.energy - dt * (0.28 + this.phase * 0.05), 0, MAX_ENERGY);
+    this.spawnOrbs(dtMs);
+    this.spawnFlybys(dtMs);
+    this.updateShadowArms(dtMs);
+    this.updateOrbs(dtMs, dt);
+    this.updateFlybys(dtMs, dt);
+    this.updatePhase();
+
+    if (this.energy <= 0) {
```

```
+      this.endRun();
+    }
+  }
+
+  applyTap(x: number, y: number): TapEvent {
+    let target: TapEvent['target'] = 'empty';
+    const flyby = this.findFlyby(x, y);
+    if (flyby) {
+      this.collectFlyby(flyby);
+      target = 'flyby';
+    } else {
+      const orb = this.findOrb(x, y, ORB_CAPTURE_RADIUS);
+      if (orb) {
+        orb.frozenMs = Math.max(orb.frozenMs, 520);
+        this.score += 8 + this.phase;
+        target = 'orb';
+      } else if (this.stunArm(x, y)) {
+        target = 'arm';
+      }
+    }
+
+    const event: TapEvent = { t: Math.round(this.timeMs), x: Math.round(x), y: Math.round(y), target };
+    this.tapEvents.push(event);
+    return event;
+  }
+
+  applySwipe(x1: number, y1: number, x2: number, y2: number): SwipeEvent {
+    const orb = this.findOrbOnSegment(x1, y1, x2, y2);
+    const target: SwipeEvent['target'] = orb ? 'orb' : 'empty';
+    if (orb) {
+      this.captureOrb(orb);
+    }
+    const event: SwipeEvent = {
+      t: Math.round(this.timeMs),
+      x1: Math.round(x1),
+      y1: Math.round(y1),
+      x2: Math.round(x2),
+      y2: Math.round(y2),
+      target
+    };
+    this.swipeEvents.push(event);
+    return event;
+  }
+
+  forceEnd(): void {
+    this.energy = 0;
+    this.endRun();
+  }
+
+  getSnapshot(): SimulationSnapshot {
+    return {
+      timeMs: Math.round(this.timeMs),
+      score: this.score,
+      energy: this.energy,
+      energyCaptured: this.energyCaptured,
+      streak: this.streak,
+      maxStreak: this.maxStreak,
+      phase: this.phase,
+      ended: this.ended,
+      collapseT: clamp(this.collapseMs / 2200, 0, 1),
+      orbs: this.orbs,
+      flybys: this.flybys,
+      shadowArms: this.shadowArms
+    };
+  }
+
+  getReplayPayload(): ReplayPayload {
+    return {
+      version: 1,
+      seed: this.seed,
+      startedAt: this.startedAt,
+      survivalMs: Math.round(this.ended ? this.timeMs - this.collapseMs : this.timeMs),
+      score: this.score,
+      energyCaptured: this.energyCaptured,
+      maxStreak: this.maxStreak,
+      tapEvents: this.tapEvents.map((event) => ({ ...event })),
+      swipeEvents: this.swipeEvents.map((event) => ({ ...event })),
+      phaseTransitions: this.phaseTransitions.map((event) => ({ ...event }))
+    };
+  }
+
+  runInputs(inputs: TimedInput[], untilMs: number): ReplayPayload {
+    let inputIndex = 0;
+    const ordered = [...inputs].sort((a, b) => a.t - b.t);
+    while (this.timeMs < untilMs && !this.ended) {
+      while (inputIndex < ordered.length && ordered[inputIndex].t <= this.timeMs) {
+        const input = ordered[inputIndex];
+        if (input.kind === 'tap') {
+          this.applyTap(input.x, input.y);
+        } else {
+          this.applySwipe(input.x, input.y, input.x2 ?? input.x, input.y2 ?? input.y);
+        }
```

```
+        inputIndex += 1;
+      }
+      this.step(1000 / 60);
+    }
+    return this.getReplayPayload();
+  }
+
+  private initPools(): void {
+    this.orbs.length = 0;
+    this.flybys.length = 0;
+    this.shadowArms.length = 0;
+    for (let index = 0; index < ORB_POOL_SIZE; index += 1) {
+      this.orbs.push(this.makeInactiveOrb(index));
+    }
+    for (let index = 0; index < FLYBY_POOL_SIZE; index += 1) {
+      this.flybys.push(this.makeInactiveFlyby(index));
+    }
+    for (let index = 0; index < 3; index += 1) {
+      this.shadowArms.push({
+        angle: index * ((Math.PI * 2) / 3),
+        stunMs: 0,
+        intensity: 0
+      });
+    }
+  }
+
+  private makeInactiveOrb(index: number): OrbState {
+    return {
+      active: false,
+      captured: false,
+      id: -index,
+      x: 0,
+      y: 0,
+      vx: 0,
+      vy: 0,
+      radius: 26,
+      ageMs: 0,
+      frozenMs: 0,
+      energy: 1,
+      tetherPhase: 0
+    };
+  }
+
+  private makeInactiveFlyby(index: number): FlybyState {
+    return {
+      active: false,
+      id: -index,
+      x: 0,
+      y: 0,
+      vx: 0,
+      vy: 0,
+      radius: 24,
+      ageMs: 0
+    };
+  }
+
+  private spawnOrbs(dtMs: number): void {
+    this.orbSpawnMs -= dtMs;
+    if (this.orbSpawnMs > 0) {
+      return;
+    }
+    this.orbSpawnMs = clamp(1020 - this.phase * 120 - this.timeMs * 0.006, 430, 1100);
+    const orb = this.orbs.find((candidate) => !candidate.active);
+    if (!orb) {
+      return;
+    }
+
+    const edge = Math.floor(this.rng() * 4);
+    const margin = 96;
+    let x = margin + this.rng() * (WORLD_WIDTH - margin * 2);
+    let y = margin + this.rng() * (WORLD_HEIGHT - margin * 2);
+    if (edge === 0) {
+      y = margin;
+    } else if (edge === 1) {
+      x = WORLD_WIDTH - margin;
+    } else if (edge === 2) {
+      y = WORLD_HEIGHT - margin * 1.9;
+    } else {
+      x = margin;
+    }
+    const angle = Math.atan2(y - BLACK_HOLE_Y, x - BLACK_HOLE_X);
+    const tangentSign = this.rng() > 0.5 ? 1 : -1;
+    const tangent = angle + (Math.PI / 2) * tangentSign + (this.rng() - 0.5) * 0.45;
+    const speed = 76 + this.rng() * 48 + this.phase * 12;
+    const inwardX = BLACK_HOLE_X - x;
+    const inwardY = BLACK_HOLE_Y - y;
+    const inwardLength = Math.max(1, Math.hypot(inwardX, inwardY));
+
+    orb.active = true;
+    orb.captured = false;
+    orb.id = this.nextOrbId;
+    this.nextOrbId += 1;
```

```
+     orb.x = x;
+     orb.y = y;
+     orb.vx = Math.cos(tangent) * speed + (inwardX / inwardLength) * 42;
+     orb.vy = Math.sin(tangent) * speed + (inwardY / inwardLength) * 42;
+     orb.radius = 24 + this.rng() * 13;
+     orb.ageMs = 0;
+     orb.frozenMs = 0;
+     orb.energy = 2 + this.rng() * 2.8;
+     orb.tetherPhase = this.rng() * Math.PI * 2;
+   }
+
+   private spawnFlybys(dtMs: number): void {
+     this.flybySpawnMs -= dtMs;
+     if (this.flybySpawnMs > 0) {
+       return;
+     }
+     this.flybySpawnMs = 4700 + this.rng() * 1900 - this.phase * 320;
+     const flyby = this.flybys.find((candidate) => !candidate.active);
+     if (!flyby) {
+       return;
+     }
+     const fromLeft = this.rng() > 0.5;
+     const y = 240 + this.rng() * (WORLD_HEIGHT - 620);
+     const speed = 760 + this.rng() * 240;
+     flyby.active = true;
+     flyby.id = this.nextFlybyId;
+     this.nextFlybyId += 1;
+     flyby.x = fromLeft ? -80 : WORLD_WIDTH + 80;
+     flyby.y = y;
+     flyby.vx = fromLeft ? speed : -speed;
+     flyby.vy = (this.rng() - 0.5) * 110;
+     flyby.radius = 22 + this.rng() * 9;
+     flyby.ageMs = 0;
+   }
+
+   private updateShadowArms(dtMs: number): void {
+     const dt = dtMs / 1000;
+     for (let index = 0; index < this.shadowArms.length; index += 1) {
+       const arm = this.shadowArms[index];
+       arm.angle += dt * (0.12 + this.phase * 0.045) * (index % 2 === 0 ? 1 : -1);
+       arm.stunMs = Math.max(0, arm.stunMs - dtMs);
+       arm.intensity = arm.stunMs > 0 ? 0.25 : clamp((this.phase - 1) / 3, 0, 1);
+     }
+   }
+
+   private updateOrbs(dtMs: number, dt: number): void {
+     for (const orb of this.orbs) {
+       if (!orb.active) {
+         continue;
+       }
+
+       orb.ageMs += dtMs;
+       if (orb.captured) {
+         const captureT = clamp(dt * 8, 0, 1);
+         orb.x += (BLACK_HOLE_X - orb.x) * captureT;
+         orb.y += (WORLD_HEIGHT - 112 - orb.y) * captureT;
+         orb.radius *= 1 + dt * 0.8;
+         if (orb.ageMs > 460) {
+           orb.active = false;
+         }
+         continue;
+       }
+
+       if (orb.frozenMs > 0) {
+         orb.frozenMs = Math.max(0, orb.frozenMs - dtMs);
+       } else {
+         const dx = BLACK_HOLE_X - orb.x;
+         const dy = BLACK_HOLE_Y - orb.y;
+         const radiusSq = dx * dx + dy * dy;
+         const radius = Math.max(48, Math.sqrt(radiusSq));
+         const gravity = (this.phase >= 3 ? 420000 : 190000) / radiusSq;
+         orb.vx += (dx / radius) * gravity * dt;
+         orb.vy += (dy / radius) * gravity * dt;
+         this.applyArmInfluence(orb, dt);
+         orb.x += orb.vx * dt;
+         orb.y += orb.vy * dt;
+       }
+
+       const distanceToHole = distanceSquared(orb.x, orb.y, BLACK_HOLE_X, BLACK_HOLE_Y);
+       if (distanceToHole < EVENT_HORIZON_RADIUS * EVENT_HORIZON_RADIUS) {
+         this.missOrb(orb);
+       } else if (
+         orb.x < -240 ||
+         orb.x > WORLD_WIDTH + 240 ||
+         orb.y < -260 ||
+         orb.y > WORLD_HEIGHT + 260 ||
+         orb.ageMs > 17500
+       ) {
+         this.missOrb(orb);
+       }
+     }
+   }
```

```
+
+  private updateFlybys(dtMs: number, dt: number): void {
+    for (const flyby of this.flybys) {
+      if (!flyby.active) {
+        continue;
+      }
+      flyby.ageMs += dtMs;
+      flyby.x += flyby.vx * dt;
+      flyby.y += flyby.vy * dt;
+      if (
+        flyby.x < -180 ||
+        flyby.x > WORLD_WIDTH + 180 ||
+        flyby.y < -160 ||
+        flyby.y > WORLD_HEIGHT + 160 ||
+        flyby.ageMs > 4500
+      ) {
+        flyby.active = false;
+      }
+    }
+  }
+
+  private applyArmInfluence(orb: OrbState, dt: number): void {
+    if (this.phase < 2) {
+      return;
+    }
+    for (const arm of this.shadowArms) {
+      if (arm.stunMs > 0) {
+        continue;
+      }
+      const endX = BLACK_HOLE_X + Math.cos(arm.angle) * ARM_LENGTH;
+      const endY = BLACK_HOLE_Y + Math.sin(arm.angle) * ARM_LENGTH;
+      const distance = distanceToSegmentSquared(orb.x, orb.y, BLACK_HOLE_X, BLACK_HOLE_Y, endX, endY);
+      if (distance > (ARM_HIT_WIDTH + orb.radius) * (ARM_HIT_WIDTH + orb.radius)) {
+        continue;
+      }
+      const normalX = Math.cos(arm.angle + Math.PI / 2);
+      const normalY = Math.sin(arm.angle + Math.PI / 2);
+      const push = this.phase >= 4 ? -130 : 92;
+      orb.vx += normalX * push * dt;
+      orb.vy += normalY * push * dt;
+      if (this.phase >= 4) {
+        orb.vx += (BLACK_HOLE_X - orb.x) * 0.17 * dt;
+        orb.vy += (BLACK_HOLE_Y - orb.y) * 0.17 * dt;
+      }
+    }
+  }
+
+  private updatePhase(): void {
+    const nextPhase: GamePhase =
+      this.energy > 64 ? 1 : this.energy > 39 ? 2 : this.energy > 18 ? 3 : 4;
+    if (nextPhase !== this.phase) {
+      this.phase = nextPhase;
+      this.phaseTransitions.push({
+        t: Math.round(this.timeMs),
+        phase: this.phase,
+        energy: Math.round(this.energy * 10) / 10
+      });
+    }
+  }
+
+  private findOrb(x: number, y: number, radius: number): OrbState | undefined {
+    let best: OrbState | undefined;
+    let bestDistance = Number.POSITIVE_INFINITY;
+    for (const orb of this.orbs) {
+      if (!orb.active || orb.captured) {
+        continue;
+      }
+      const hitRadius = radius + orb.radius;
+      const dist = distanceSquared(x, y, orb.x, orb.y);
+      if (dist < hitRadius * hitRadius && dist < bestDistance) {
+        best = orb;
+        bestDistance = dist;
+      }
+    }
+    return best;
+  }
+
+  private findOrbOnSegment(x1: number, y1: number, x2: number, y2:
```

[Diff truncated for PDF practicality. package-lock and binary images are represented in the stat above.]